

Math 242, S16
Quiz 2

Name: Answer
SSN: xxx-xx-

Instructions: There are totally 2 problems. You must show all of your work to receive a credit for a correct answer.

Problem 1. (5 points) Find a general solution (explicit if convenient) for the first-order differential equations $\frac{dy}{dx} = 2x \sec(y)$.

$$\frac{dy}{dx} = 2x \sec(y)$$

$$\frac{dy}{dx} = \frac{2x}{\cos y}$$

$$\cos y \, dy = 2x \, dx$$

$$\int \cos y \, dy = \int 2x \, dx$$

$$\sin y = x^2 + C$$

$$y = \arcsin(x^2 + C)$$

Problem 2. (5 points) Find the explicit particular solution for the initial value problem $\frac{dy}{dx} = y - 2xy$, $y(1) = -4$.

$$\frac{dy}{dx} = y(1-2x)$$

$$\frac{1}{y} dy = (1-2x) dx$$

$$\int \frac{1}{y} dy = \int (1-2x) dx$$

$$\ln|y| = x - x^2 + C$$

$$|y| = e^{x-x^2+C}$$

$$= Ae^{x-x^2} \quad (A > 0)$$

$$y = Ae^{x-x^2} \text{ or } y = -Ae^{x-x^2}$$

By $y(1) = -4$, we know
 $y = -Ae^{x-x^2}$

and $-Ae^{1-1^2} = -4$

$$A = 4$$

Thus

$$y = -4e^{x-x^2}$$